



# **Gastrointestinal Pharmacology**

## **Drugs used for treatment acid related disorders**

Dr. Begard O. Berzinji PhD Pharmacology  
HMU Pharmacology Department

# Obejctives

- Regulation of gastric acid secretion
- Classiffication of drugs used in peptic ulcer disease
- Mechanison of action, uses and adverse effects, drug interactions of
  - ❖ H2 antagonist
  - ❖ Proton pump inhibitors
  - ❖ antacids
  - ❖ Cytoprotective agents

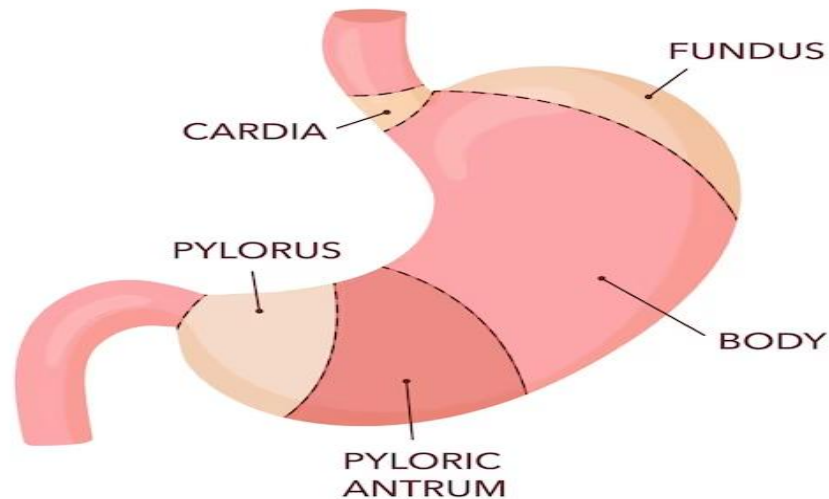
## Regulation of Stomach Acid Release

- **Parietal cells:** (located in body and fundus of stomach) are responsible for secretion of hydrochloric acid and intrinsic factor (required for vitamin B12 absorption).

The parietal cell contains receptors for gastrin, acetylcholine (muscarinic, M3), and histamine (H2).

Neural release of acetylcholine and gastrin stimulates release of acid from the parietal cells by activating the H<sup>+</sup>/K<sup>+</sup> (ATPase) pump via a Ca<sup>2+</sup> dependent protein kinase

- **G cell:** a type of endocrine cell (in the gastric antrum and duodenum )that secret gastrin.



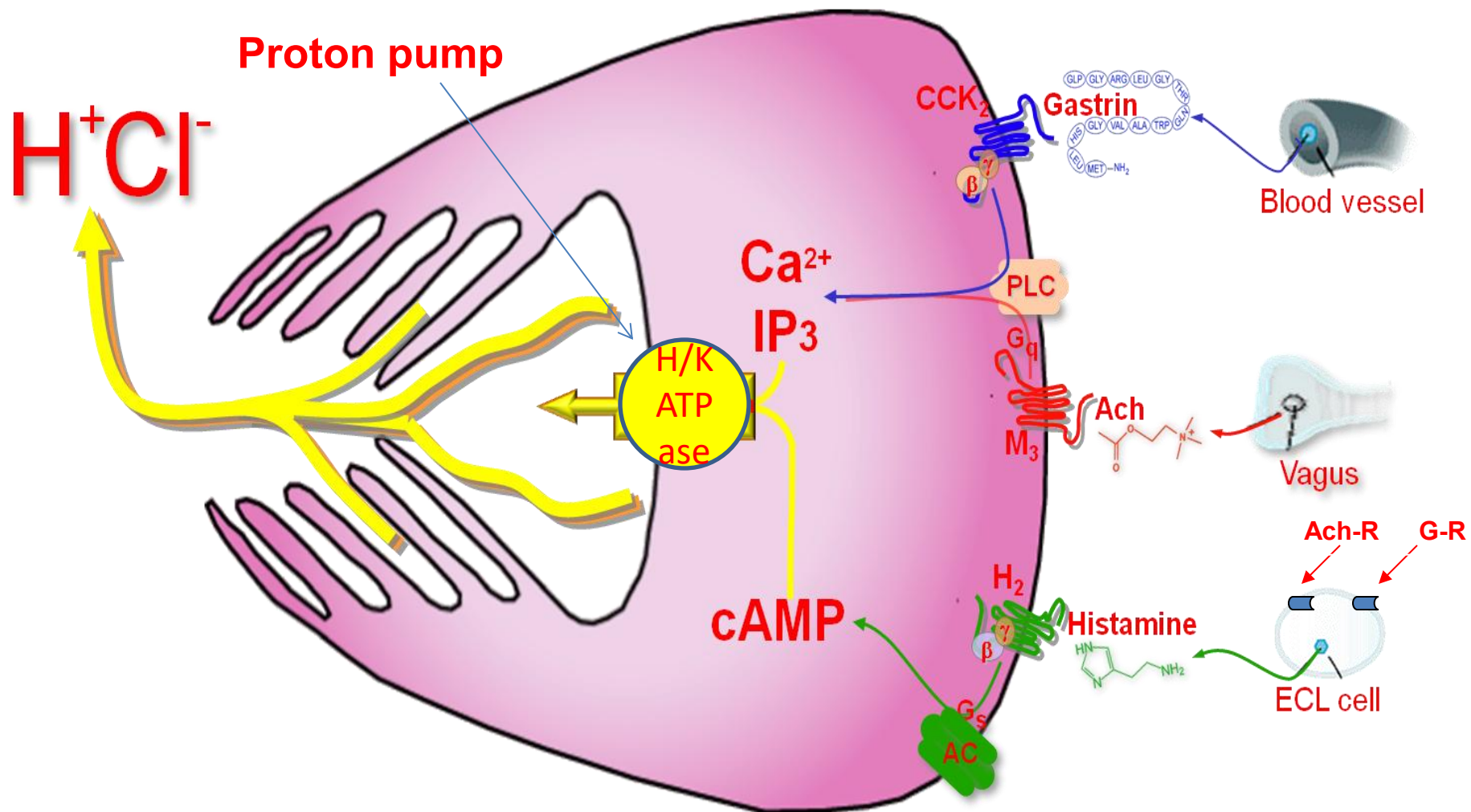
- **The enterochromaffin-like (ECL) cells:** in the fundus of the stomach produce, store and secrete histamine in response to acetylcholine and gastrin.

Gastrin and acetylcholine receptors are found on ECL cells as well as on parietal cells.

Acetylcholine and gastrin induce histamine release from enterochromaffin-like cells, which then stimulate the  $H^+/K^+$  ATPase pump (through stimulation of  $H_2$  receptors) on parietal cells.

In contrast to acetylcholine and gastrin, H<sub>2</sub> receptors stimulate the H<sup>+</sup>/K<sup>+</sup> ATPase pump via activation of adenylate cyclase to form cAMP and stimulate protein kinase A. Protein kinase A induces fusion of tubulovesicles within the canalicular membrane of the parietal cell to release protons into the gastric lumen.

**Basal Acid Output (BAO):** Is the minimum amount of gastric hydrochloric acid produced by an individual in a given period (Normal adult volume is 2 to 5 mEq/hr)



**Mechanism of acid production**

## **Mucus production:**

- Produced by mucus-secreting cells
- Also produce bicarbonate, which becomes trapped in the mucus layer
- Prostaglandins E2 and I2 inhibit acid secretion (by inhibition of adenyl cyclase), but stimulate mucus and bicarbonate secretion, and dilate mucosal blood vessels.



## **Acid Related Diseases**

The majority of upper GI disorders are due to an increase in acid secretion which include;

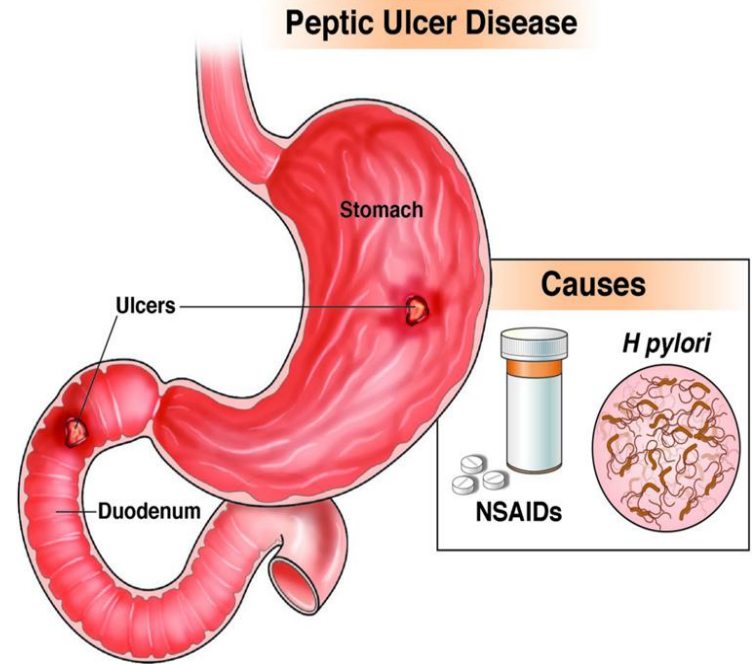
- Dyspepsia
- Gastroesophageal reflux disease (GERD)
- Peptic ulcer disease (PUD)
- Upper GI bleeding.

# Peptic Ulcer Diseases

Peptic ulcer (especially duodenal) is a chronic remitting and relapsing disease lasting several years.

The goals of antiulcer therapy are:

- Relief of pain
- Ulcer healing
- Prevention of complications (bleeding, perforation)
- Prevention of relapse



Peptic ulcer disease affects approximately 10% of the population of western countries. The incidence of duodenal ulcer (DU) is four to five times higher than that of gastric ulcer (GU).

Although the pathogenesis of peptic ulcer disease is not fully understood, several major causative factors are recognized:

- Nonsteroidal anti-inflammatory drug (NSAID) use
- Infection with *Helicobacter pylori*,
- Increased hydrochloric acid secretion, and
- Inadequate mucosal defense against gastric acid.

# Drugs used to treat peptic ulcer disease

## A. Antimicrobial agents

Patients with peptic ulcer disease (both duodenal and gastric ulcers) who are infected with *H. pylori* require antimicrobial treatment.

Successful eradication of *H. pylori* (80-90 percent) is possible with various combinations of antimicrobial drugs, either:

- **Triple therapy:** consisting of a PPI with either metronidazole or amoxicillin plus clarithromycin  
**or**

- **Quadruple therapy:** bismuth subsalicylate plus metronidazole plus tetracycline plus a PPI,

They are administered for a 2-week course. This usually results in a 90 percent or greater eradication rate.

## **B. PPI (proton pump inhibitors):**

PPIs bind to the  $H^+/K^+-ATPase$  enzyme system (proton pump) of the parietal cell and effectively inhibit acid secretion. e.g: omeprazole, lansoprazole, rabeprazole, pantoprazole, and esomeprazole.

These agents are prodrugs, absorbed and transported to the parietal cell canaliculus converted to the active form, reacts with a cysteine residue of the  $H^+/K^+-ATPase$ , forming a stable covalent bond and irreversibly inactivate the proton pump (which is the final step in the secretion of gastric acid).

## Pharmacokinetics:

- They are effective orally,
- bioavailability is decreased 50% by food therefore should be administered approximately 1 hour before a meal, so that the peak serum concentration coincides with the maximal activity of proton pump secretion.
- They have a short serum half-life of about 1.5 hours, but the duration of acid inhibition lasts up to 24 hours.

due to the irreversible inactivation of the proton pump, about 18 hours needed for the enzyme to be resynthesized.

## Therapeutic uses:

- PPIs are superior over the H<sub>2</sub> antagonists for suppressing acid production and healing peptic ulcers
- Long-term treatment of Zollinger-Ellison syndrome (pathologic conditions in which a gastrin-producing tumor causes hyper secretion of HCl).
- They are approved for the treatment of GERD.

## Adverse effects:

The PPIs are generally well tolerated.

normally acid is important in releasing vitamin B<sub>12</sub> from food, but long-term treatment with PPIs may lead to

**subnormal B<sub>12</sub> levels.**



## C. H<sub>2</sub>receptor antagonists

- Prescription of H<sub>2</sub>-blockers has declined markedly with the advent of proton pump inhibitors.
- Four H<sub>2</sub>antagonists are in clinical use: **cimetidine**, **ranitidine**, **famotidine**, and **nizatidine**. They potently inhibit (greater than 90%) basal, food-stimulated, and nocturnal secretion of gastric acid.

- **Pharmacodynamics:**

They act selectively and competitively on H<sub>2</sub> receptors and they are fully reversible. These agents reduce the intracellular concentrations of cyclic adenosine monophosphate (cAMP) and, thereby, secretion of gastric acid.

cimetidine, ranitidine, and famotidine undergo first-pass hepatic metabolism resulting in a bioavailability of approximately 50%. Nizatidine has little first-pass and a bioavailability of almost 100%.

- H<sub>2</sub>antagonists are cleared by liver and kidney. but nizatidine is eliminated principally by the kidney.
- Dose reduction of H<sub>2</sub>antagonists is required in patients with moderate to severe renal (and possibly severe hepatic) insufficiency and in elderly.
- **cimetidine inhibits cytochrome P450** and thus potentiate the action of several drugs (for example, warfarin, diazepam, phenytoin, quinidine, carbamazepine, theophylline, and imipramine) resulting in serious adverse clinical effects.

## **Therapeutic uses:**

- they are used for peptic ulcers,
- Acute stress ulcer they are usually injected intravenously
- and used in Gastroesophageal reflux disease(GERD)  
However,PPIs are now used preferentially in the treatment of this disorder.

## Adverse Effects:

- H2 antagonists are extremely safe drugs.
- Side effects include diarrhea, headache, fatigue, muscular pain, and constipation. Other CNS effects (confusion, hallucinations)
- **Cimetidine: can also cause male sexual dysfunction and gynecomastia.**
- Ranitidine has minimal side effects

## D. Prostaglandins:

**PGE2** and **PGI2** are produced in the gastric mucosa and appear to serve a protective role by inhibiting acid secretion and promoting mucus as well as  $\text{HCO}_3^-$  secretion. In addition, PGs inhibit gastrin release, increase mucosal blood flow and have “cytoprotective” action.

Also they have ability to reinforce the mucus layer covering gastric and duodenal mucosa which is buffered by  $\text{HCO}_3^-$  secreted into this layer by the underlying epithelial cells.

- **Misoprostol:** Natural PGs have very short  $t_{1/2}$ . Misoprostol is synthetic PGE1 derivative, binds to a prostaglandin receptor on parietal cells, reducing histamine-stimulated cAMP production and causing modest acid inhibition and stimulates secretion of mucus and bicarbonate (cytoprotective effect).

However, reduction in 24-hour acid inhibition is less than H2 blockers because of shorter duration of action (~3 hr.) Ulcer healing rates comparable to cimetidine have been obtained in 4–8 weeks.

Misoprostol with PPIs, are used for prevention of gastric ulcers induced by NSAIDs . It is less effective than H2 antagonists and the PPIs for acute treatment of peptic ulcers.

misoprostol produces uterine contractions and is contraindicated during pregnancy. Dose-related diarrhea and nausea are the most common adverse effects that limit the use of this agent.

### **E. Antimuscarinic agents (anticholinergic agents)**

A cholinergic antagonist, such as pirenzepine (It is a selective M1 blocker) or dicyclomine( blocking M1, M3, and M2 receptors) , can be used in the management of peptic ulcer disease and Zollinger-Ellison syndrome, particularly in patients who are refractory to standard therapies.

However, its many side effects (for example, cardiac arrhythmias, dry mouth, constipation, and urinary retention) limit its use.

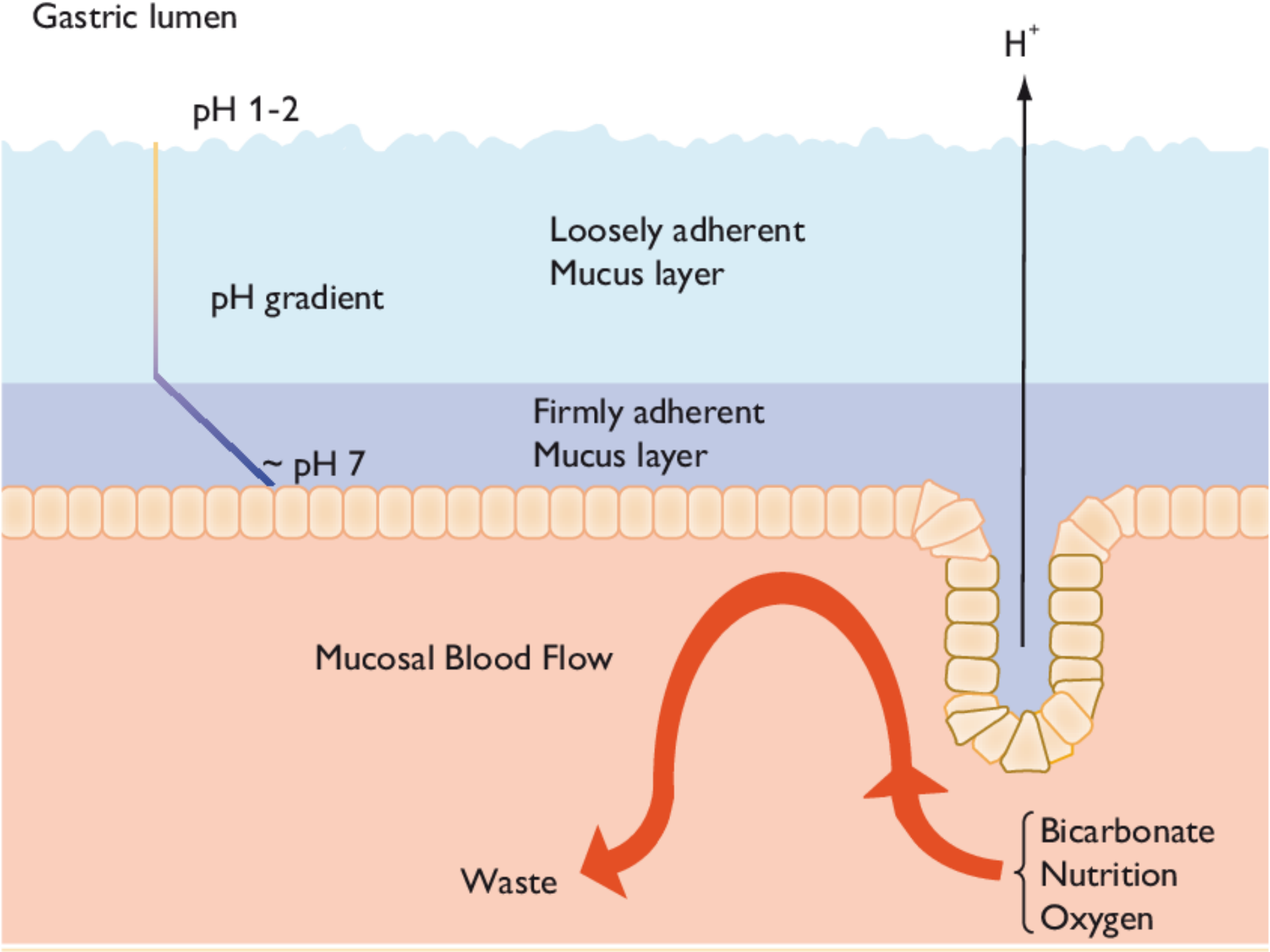


## **F. Mucosal Protective Agents**

Both mucus and epithelial cell-cell tight junctions restrict back diffusion of acid and pepsin.

Epithelial bicarbonate secretion establishes a pH gradient within the mucous layer( in which the pH ranges from 7 at the mucosal surface to 1–2 in the gastric lumen). Blood flow carries bicarbonate and vital nutrients to surface cells.

Mucosal prostaglandins appear to be important in stimulating mucus and bicarbonate secretion and mucosal blood flow.



A number of agents that potentiate these mucosal defensive mechanisms are known as cytoprotective compounds

- **Sucralfate:** is a salt of sucrose complexed to sulfated aluminum hydroxide. In water or acidic solutions, it forms a viscous, tenacious paste that binds selectively to ulcers or erosions for up to 6 hours. forming a physical barrier that restricts further gastric damage and stimulates mucosal prostaglandin and bicarbonate secretion.

Because it requires an acidic pH for activation, sucralfate should not be administered with H<sub>2</sub> antagonists or antacids. It heals duodenal ulcers, but does not prevent NSAID-induced ulcers, nor does it heal gastric ulcers.

- **Bismuth subsalicylate:** this compound effective in healing peptic ulcers. In addition to their antimicrobial actions, they coat the ulcer base inhibit the activity of pepsin, increase secretion of mucus.

**Unwanted effects** include nausea and vomiting, and blackening of the tongue and feces.

## G. Antacids:

Antacids are weak bases that react with gastric acid to form water and a salt and cause:

1. Partial neutralization of gastric acid within the gastric lumen.
2. Inhibition of pepsin activity within the gastric lumen, secondary to decrease in gastric acid, because pepsin is inactive at a pH greater than 4.

Antacids are very quick acting and modestly elevate intra-gastric pH within minutes, but their duration of action is short (about 30-60 minutes on an empty stomach).

Commonly used antacids are:

## **Systemic Antacids:**

**NaHCO<sub>3</sub>:** sodium bicarbonate is a potent acid neutralizer (pH may rise above 7). It is water soluble and absorbed systemically at high doses which may cause:

**(a)** Systemic alkalosis

**(b)** Sodium load that may worsen edema and congestive heart failure (CHF).

**(c)** Produces CO<sub>2</sub> in stomach → distention, discomfort, belching, risk of ulcer perforation.

**Use of NaHCO<sub>3</sub>:** is restricted to casual treatment of heartburn. It provides quick symptomatic relief. Other uses are to alkalinize urine and to treat acidosis.

## Non-Systemic Antacids:

These are insoluble and poorly absorbed basic compounds; react in stomach to form the corresponding chloride salt. The chloride salt again reacts with the intestinal bicarbonate so that  $\text{HCO}_3^-$  is not spared for absorption—no acid-base disturbance occurs.

**Mg(OH)<sub>2</sub>:** Can cause diarrhea and loose stools when used alone, also it may cause Hyporeflexia, hypotension, and cardiac arrest

**Al(OH)<sub>3</sub>:** The  $\text{Al}^{3+}$  ions relax smooth muscle thus, it delays gastric emptying and frequently causes constipation when used alone.

Combination Mg(OH)<sub>2</sub> and Al(OH)<sub>3</sub> may annul each other's action and bowel movement may be least affected.

The  $\text{Al}^{3+}$  ions binds phosphate in the intestine and prevents its absorption—hypophosphatemia occurs on regular use. This may:

**(a)** cause osteomalacia

**(b)** be used therapeutically in hyperphosphatemia and phosphate stones.

Small amount of  $\text{Al}^{3+}$  that is absorbed is excreted by kidney. This is impaired in renal failure—aluminium toxicity (encephalopathy, osteoporosis) can occur.



**CaCO<sub>3</sub>:** It is a potent and rapidly acting acid neutralizer its a commonly used preparation. Reacts with HCl in the stomach and liberates CO<sub>2</sub> but at a slower rate than NaHCO<sub>3</sub>. It can cause distention and discomfort.

### **Contraindications:**

1. Hypertension, heart failure
2. Renal disease

### **Drug interaction:**

- **Alkalinization of gastric contents can change the bioavailability** of a large number of drugs: iron, tetracyclines, digoxin, prednisolone, propranolol, quinidine, antimuscarinics

- Ions in antacids can **chelate** with drugs, forming insoluble complexes e.g. **Quinolones** with  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ , or  $\text{Al}^{3+}$ -**Tetracyclines** with  $\text{Ca}^{2+}$  .
- Also, **alkalinization of urine alters the urinary excretion** of many drugs, it increases excretion of **phenobarbital, methotrexate, salicylate**